Chapter 3

(3.1) $\langle \vec{x}, \vec{y} \rangle = x_1 y_1 - (x_1 y_2 + x_2 y_1) + 2 x_2 y_2$

First we need to show that this is a bilinear mapping

$\Omega(\lambda \vec{x}, \psi \vec{y}) = \lambda \Omega(\vec{x}, \psi \vec{y}) = \lambda \psi \, \Omega(\vec{x}, \vec{y})$

$\langle \lambda \vec{x}, \psi \vec{y} \rangle = \lambda x_1 \psi y_1 - (\lambda x_1 \psi y_2 + \lambda x_2 \psi y_1) + 2 \lambda x_2 \psi y_2$

$= \lambda \psi x_1 y_1 - \lambda \psi (x_1 y_2 + x_2 y_1) + \lambda \psi 2 x_2 y_2$

$= \lambda \psi \left[x_1 y_1 - (x_1 y_2 + x_2 y_1) + 2 x_2 y_2 \right]$

$= \lambda \psi \langle \vec{x}, \vec{y} \rangle$ ✓

Now check if it is symmetric

$\langle y, x \rangle = y_1 x_1 - (y_1 x_2 + y_2 x_1) + 2 y_2 x_2 =$

$= x_1 y_1 - (x_1 y_2 + x_2 y_1) + 2 x_2 y_2 = \langle x, y \rangle$ ✓

Now check if it is positive definite for $\vec{x}, \vec{y} \neq \vec{0}$ and $\vec{x} = \vec{y}$

$\langle x, x \rangle = x_1 x_1 - (x_1 x_2 + x_2 x_1) + 2 x_2 x_2 =$

$= x_1^2 - 2 x_1 x_2 + x_2^2 + x_2^2 =$

$= (x_1 - x_2)^2 + x_2^2$

If $\vec{x} = \vec{0}$ we see that $\langle \vec{x}, \vec{x} \rangle = 0$ ✓

For any other x, $(x_1 - x_2)^2 + x_2^2 > 0$

So this is a valid inner product.

3.2

$$\langle x, y\rangle = x^T \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} y = [x_1 \ x_2] \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} =$$

$$= [x_1 \ x_2] \begin{bmatrix} 2y_1 \\ y_1 + 2y_2 \end{bmatrix} = 2x_1y_1 + x_2y_1 + 2x_2y_2$$

$$\langle y, x\rangle = y^T \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} x = [y_1 \ y_2] \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} =$$

$$= [y_1 \ y_2] \begin{bmatrix} 2x_1 \\ x_1 + 2x_2 \end{bmatrix} = 2x_1y_1 + y_2x_1 + 2x_2y_2$$

We see that $\langle x, y\rangle \neq \langle y, x\rangle$ and also A itself is not symmetric, so this is not a valid inner product.

Also, this inner product is not positive definite, which again means it is not valid.

3.3

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad y = \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix} \qquad x - y = \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}$$

$$d(x,y) = \sqrt{\langle x-y, x-y\rangle} = \sqrt{\left\langle \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix} \right\rangle}$$

a)

$$\sqrt{[2 \ 3 \ 3] \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}} = \sqrt{4+9+9} = \sqrt{22}$$

b)

$$\sqrt{[2 \ 3 \ 3] \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}} = \sqrt{[7 \ 8 \ 3] \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}} = \sqrt{14+24+9} = \sqrt{47}$$

3.4

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad y = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

a) $$\cos\theta = \frac{x^T y}{\|x\| \, \|y\|} = \frac{-3}{\sqrt{5} \cdot \sqrt{2}} = -\frac{3}{\sqrt{10}} \qquad \theta = \arccos\left(-\frac{3}{\sqrt{10}}\right) = 2.8198 \text{ rad}$$

b) $$\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}} = \sqrt{\begin{bmatrix} 4 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}} = \sqrt{18}$$

$$\|y\| = \sqrt{\langle y, y \rangle} = \sqrt{\begin{bmatrix} -1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix}} = \sqrt{\begin{bmatrix} -3 & -4 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix}} = \sqrt{7}$$

$$\langle x, y \rangle = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 & 7 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} = -11$$

$$\cos\theta = \frac{\langle x, y \rangle}{\|x\| \, \|y\|} = \frac{-11}{\sqrt{126}} \qquad \theta = \arccos\left(\frac{-11}{\sqrt{126}}\right) = 2.941 \text{ rad}$$

3.5

$$U = \text{span}\left[\begin{bmatrix} 0 \\ -1 \\ 2 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} -3 \\ 4 \\ 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \\ 5 \\ 0 \\ 7 \end{bmatrix} \right] \qquad x = \begin{bmatrix} -1 \\ -9 \\ -1 \\ 4 \\ 1 \end{bmatrix}$$

$$\pi_U(x) = B\lambda$$

projection of $\vec{x}$ onto the subspace U where the projection is represented by a linear combination of basis vectors

$$\lambda_1 \vec{b_1} + \lambda_2 \vec{b_2} + \dots \lambda_n \vec{b_n} = B\lambda$$

First determine which vectors are linearly independent.

$$\begin{bmatrix} 0 & 1 & -3 & -1 \\ -1 & -3 & 4 & -3 \\ 2 & 1 & 1 & 5 \\ 0 & -1 & 2 & 0 \\ 2 & 2 & 1 & 7 \end{bmatrix} \rightsquigarrow \begin{array}{c} \begin{bmatrix} -1 & -3 & 4 & -3 \\ 0 & 1 & -3 & -1 \\ 2 & 1 & 1 & 5 \\ 0 & -1 & 2 & 0 \\ 2 & 2 & 1 & 7 \end{bmatrix} \end{array} \begin{array}{l} \\ \\ +2R_1 \\ \\ +2R_1 \end{array}$$

$$\rightsquigarrow \begin{bmatrix} -1 & -3 & 4 & -3 \\ 0 & 1 & -3 & -1 \\ 0 & -5 & 9 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & -4 & 9 & 1 \end{bmatrix} \begin{array}{l} \\ \\ +5R_2 \\ +R_2 \\ +4R_2 \end{array}$$

$$\rightsquigarrow \begin{array}{l} \\ \\ /-6 \\ /-1 \\ /-3 \end{array} \begin{bmatrix} -1 & -3 & 4 & -3 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & -6 & -6 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -3 & -3 \end{bmatrix}$$

$$\rightsquigarrow \begin{array}{l} \\ \\ \\ -R_3 \\ -R_3 \end{array} \begin{bmatrix} -1 & -3 & 4 & -3 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} -1 & -3 & 4 & -3 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Rank 3, can use the first three vectors as <u>basis</u>

So we form a basis using the first three vectors. $\left(b_1 = \begin{bmatrix} 0 \\ -1 \\ 2 \\ 0 \\ 2 \end{bmatrix}, b_2 = \begin{bmatrix} 1 \\ -3 \\ 1 \\ -1 \\ 2 \end{bmatrix}, b_3 = \begin{bmatrix} -3 \\ 4 \\ 1 \\ 2 \\ 1 \end{bmatrix} \right)$

$$\pi_U(x) = B\underbrace{(B^T B)^{-1} B^T x}_{\lambda} = B\lambda$$

Note that instead of finding the inverse, we can solve

$$B^T B \lambda = B^T x \quad \text{to find } \lambda$$

$$B^T B = \underset{3\times 5}{\begin{bmatrix} 0 & -1 & 2 & 0 & 2 \\ 1 & -3 & 1 & -1 & 2 \\ -3 & 4 & 1 & 2 & 1 \end{bmatrix}} \underset{5\times 3}{\begin{bmatrix} 0 & 1 & -3 \\ -1 & -3 & 4 \\ 2 & 1 & 1 \\ 0 & -1 & 2 \\ 2 & 2 & 1 \end{bmatrix}} = \begin{bmatrix} 9 & 9 & 0 \\ 9 & 16 & -14 \\ 0 & -14 & 31 \end{bmatrix}$$

$$B^T x = \begin{bmatrix} 0 & -1 & 2 & 0 & 2 \\ 1 & -3 & 1 & -1 & 2 \\ -3 & 4 & 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ -9 \\ -1 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 23 \\ -25 \end{bmatrix}$$

So we can now set up the following matrix

$$\left[\begin{array}{ccc|c} 9 & 9 & 0 & 9 \\ 9 & 16 & -14 & 23 \\ 0 & -14 & 31 & 25 \end{array}\right] -R_1 \rightsquigarrow \left[\begin{array}{ccc|c} 9 & 9 & 0 & 9 \\ 0 & 7 & -14 & 14 \\ 0 & -14 & 31 & -25 \end{array}\right] +2R_2$$

$$\rightsquigarrow \left[\begin{array}{ccc|c} 9 & 9 & 0 & 9 \\ 0 & 7 & -14 & 14 \\ 0 & 0 & 3 & 3 \end{array}\right] \begin{array}{c} 19 \\ 17 \\ 13 \end{array}$$

$$\rightsquigarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$x_3 = 1$

$x_2 - 2 = 2 \qquad x_2 = 4$

$x_1 + 4 = 1 \qquad x_1 = -3$

$$x = \begin{bmatrix} -3 \\ 4 \\ 1 \end{bmatrix}$$

$$\Pi_U(x) = Bx = \begin{bmatrix} 0 & 1 & -3 \\ -1 & -3 & 4 \\ 2 & 1 & 1 \\ 0 & -1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} -3 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ -6 \\ 0 \\ -6 \end{bmatrix} + \begin{bmatrix} 4 \\ -12 \\ 4 \\ -4 \\ 8 \end{bmatrix} + \begin{bmatrix} -3 \\ 4 \\ 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -5 \\ -1 \\ -2 \\ 3 \end{bmatrix}$$

b)

get $d(x, U)$

This is just the distance between x and $\Pi_U(x)$

so we need to calculate $\| x - \Pi_U(x) \|$

$$\| x - \Pi_U(x) \| = \left\| \begin{bmatrix} -1 \\ -9 \\ -1 \\ 4 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ -5 \\ -1 \\ -2 \\ 3 \end{bmatrix} \right\| = \left\| \begin{bmatrix} -2 \\ -4 \\ 0 \\ 6 \\ -2 \end{bmatrix} \right\|$$

$$= \sqrt{\left\langle \begin{bmatrix} -2 \\ -4 \\ 0 \\ 6 \\ -2 \end{bmatrix}, \begin{bmatrix} -2 \\ -4 \\ 0 \\ 6 \\ -2 \end{bmatrix} \right\rangle} = \sqrt{60}$$

3.6

$$B = \begin{bmatrix} e_1 & e_3 \end{bmatrix} \qquad A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

a)

$$\langle e_1, \overbrace{e_2 - \Pi_U(e_2)}^{\text{this vector will be orthogonal to } U} \rangle = 0 \qquad e_1^T A (e_2 - \Pi_U(e_2)) = 0$$

$$\langle e_3, e_2 - \Pi_U(e_2) \rangle = 0 \qquad e_3^T A (e_2 - \Pi_U(e_2)) = 0$$

The two equations above can be combined into

$$B^T A (x - \Pi_U(e_2)) = 0$$

We also know that $\Pi_U(e_2)$ is the projection of e_2 onto the substace U, which can be expressed as a linear combination of e_1, e_3 that form a basis of U.

$$\Pi_U(e_2) = Bx = \lambda_1 \vec{e}_1 + x_3 \vec{e}_3$$

$$B^T A (e_2 - B\lambda) = 0$$

$$B^T A e_2 - B^T A B \lambda = 0$$

$$B^T A e_2 = B^T A B \lambda$$

$$(B^T A B)^{-1} B^T A e_2 = \lambda$$

$$B^T A B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 & 0 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

In this case the inverse is easy to find since it is a diagonal matrix

$$(B^T A B)^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

$$B^T A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & -1 & 2 \end{bmatrix}$$

$$(B^T A B)^{-1} B^T A x = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} =$$

$$= \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} = x$$

So $$B\lambda = \Pi_U(e_2) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ 0 \\ -\frac{1}{2} \end{bmatrix}$$

b)

Similarly, we need to compute $\| e_2 - \Pi_U(e_2) \|$

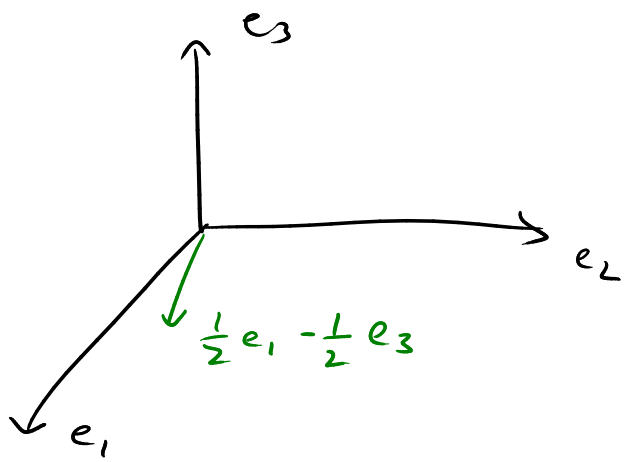
But note that this time the inner product definition is different.

$$e_2 - \Pi_U(e_2) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} \\ 0 \\ -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ 1 \\ \frac{1}{2} \end{bmatrix}$$

$$\left\| \begin{bmatrix} -\frac{1}{2} \\ 1 \\ \frac{1}{2} \end{bmatrix} \right\| = \sqrt{\left\langle \begin{bmatrix} -\frac{1}{2} \\ 1 \\ \frac{1}{2} \end{bmatrix}, \begin{bmatrix} -\frac{1}{2} \\ 1 \\ \frac{1}{2} \end{bmatrix} \right\rangle} = \sqrt{1} = 1$$

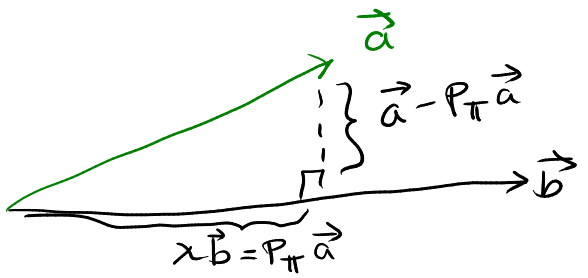
$$\begin{bmatrix} -\frac{1}{2} & 1 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} -\frac{1}{2} \\ 1 \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} -\frac{1}{2} \\ 1 \\ \frac{1}{2} \end{bmatrix} = 1$$

c)



3.7

a) The projection π can be represented with the matrix P_π



$$P_\pi \vec{a} = \lambda \vec{b}$$

$$\vec{a} - P_\pi \vec{a} = I\vec{a} - P_\pi \vec{a} = (I - P_\pi)\vec{a}$$

identity matrix
identity mapping id$_V$

By definition, applying a projection twice does not give a different result

$$P_\pi \vec{a} = P_\pi P_\pi \vec{a} = (P_\pi)^2 \vec{a}$$

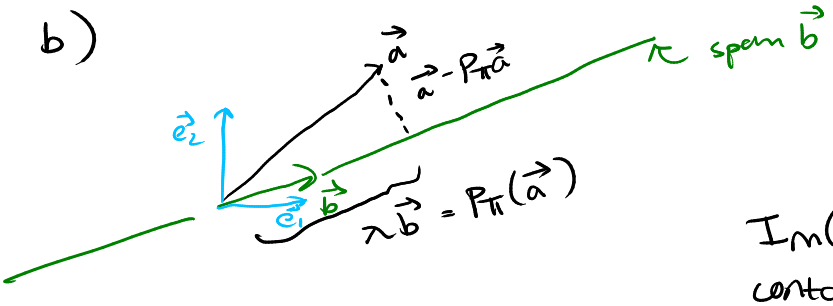
If $I - P_\pi$ is a projection matrix, then $(I - P_\pi)^2 = I - P_\pi$

Expanding $(I - P_\pi)^2$ we get:

$$(I - P_\pi)^2 = (I - 2IP_\pi + P_\pi^2) = (I - 2P_\pi + P_\pi) = I - P_\pi$$

which is only true if $P_\pi^2 = P_\pi$, i.e. P_π is a projection matrix

b)



Assume that $\dim(V) = 2$ and $\vec{a} \in V$.

$\vec{a} = a_1\vec{e}_1 + a_2\vec{e}_2$

$P_\pi\vec{a} = \lambda\vec{b}$

$Im(id_V - \pi)$ is the vector subspace containing the vectors:

$\{\vec{w} : P_\pi(\vec{w}) = 0\}$

This is the set of vectors which map to the zero vector after applying P_π, i.e. this is the $Ker(\pi)$.

$= \{t : (I-P)\vec{t} = 0\}$

Similarly $ker(id_V - \pi) = \{w : P_\pi(\vec{w}) = \lambda\vec{b}\}$, i.e this is the set of vectors which map to the zero vector after applying the $(id_V - \pi)$ to them, which is the $Im(\pi)$

$$Dim(R^2) = 2 = Dim(Im(id_V - \pi)) + Dim(Ker(id_V - \pi)) = Dim(Ker(\pi)) + Dim(Im(\pi))$$

but also applies to higher dimensions. For example, if projecting a 3D vector onto a line, then $Dim(Im(id_V - \pi)) = 2$ and $Dim(Ker(id_V - \pi)) = 1$ and $Dim(Ker(\pi)) = 2$, $Dim(Im(\pi)) = 1$

3.8

$$\vec{b}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \vec{b}_2 = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$$

$$\vec{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \boxed{\vec{c}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}}$$

(note we can first find an orthogonal basis and then scale the basis vectors to be unit vectors).

$$\|\vec{u}_1\| = \sqrt{3}$$

$$\vec{u}_2 = \vec{b}_2 - \frac{\vec{u}_1 \cdot \vec{u}_1^T}{\|\vec{u}_1\|^2} \vec{b}_2 = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} =$$

$$= \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{4}{3} \\ \frac{5}{3} \\ -\frac{1}{3} \end{bmatrix}$$

$$\vec{c}_2 = \frac{\vec{u}_2}{\|\vec{u}_2\|} = \frac{3}{\sqrt{42}} \cdot \begin{bmatrix} -\frac{4}{3} \\ \frac{5}{3} \\ -\frac{1}{3} \end{bmatrix} = \boxed{\frac{1}{\sqrt{42}} \begin{bmatrix} -4 \\ 5 \\ -1 \end{bmatrix}}$$

$$\|\vec{u}_2\| = \sqrt{\frac{16}{9} + \frac{25}{9} + \frac{1}{9}} = \sqrt{\frac{42}{9}} = \frac{\sqrt{42}}{3}$$

3.9

$$\sum_{i=1}^{n} x_i = 1 \qquad |\vec{x} \cdot \vec{y}| \leq \|x\| \, \|y\| \quad \text{and} \quad (\vec{x} \cdot \vec{y})^2 \leq \|x\|^2 \|y\|^2$$

a)

Choosing $\vec{y} = \begin{bmatrix} \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} \end{bmatrix}^T$

$$\vec{x} \cdot \vec{y} = \sum_{i=1}^{n} x_i \cdot \frac{1}{n} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

this is equal to 1 by def.

$$(\vec{x} \cdot \vec{y})^2 = \frac{1}{n^2} \left(\sum_{i=1}^{n} x_i \right)^2 = \frac{1}{n^2}$$

$$\|x\|^2 = \sum_{i=1}^{n} x_i^2 \qquad \|y\|^2 = \sum_{i=1}^{n} \left(\frac{1}{n}\right)^2 = \left(\frac{1}{n}\right)^2 \cdot \sum_{i=1}^{n} 1 = \frac{1}{n^2} \cdot n = \frac{1}{n}$$

Plugging in $(\vec{x} \cdot \vec{y})^2 = \frac{1}{n^2}$. and $\|x\|^2 = \sum_{i=1}^{n} x_i^2$ and $\|y\|^2 = \frac{1}{n}$

into the Cauchy-Schwarz inequality we get.

$$\frac{1}{n^2} \leq \frac{1}{n} \sum_{i=1}^{n} x_i^2$$

$$\frac{1}{n} \leq \sum_{i=1}^{n} x_i^2$$

b)

$$\sum_{i=1}^{n} x_i = 1$$

$$\vec{x} = \begin{bmatrix} \sqrt{x_1} & \sqrt{x_2} & \cdots & \sqrt{x_n} \end{bmatrix}^T$$

$$\vec{y} = \begin{bmatrix} \frac{1}{\sqrt{x_1}} & \frac{1}{\sqrt{x_2}} & \cdots & \frac{1}{\sqrt{x_n}} \end{bmatrix}^T$$

$$\vec{x} \cdot \vec{y} = \sum_{i=1}^{n} \sqrt{x_i} \cdot \frac{1}{\sqrt{x_i}} = \sum_{i=1}^{n} 1 = n$$

$$\|x\|^2 = \sum_{i=1}^{n} (\sqrt{x_i})^2 = \sum_{i=1}^{n} x_i = 1 \leftarrow \text{by definition}$$

$$\|y\|^2 = \sum_{i=1}^{n} \left(\frac{1}{\sqrt{x_i}}\right)^2 = \sum_{i=1}^{n} \frac{1}{x_i}$$

$$(\vec{x} \cdot \vec{y})^2 = n^2$$

Applying Cauchy-Schwartz inequality we get

$$(\vec{x} \cdot \vec{y})^2 \leq \|x\| \, \|y\|$$

$$n^2 \leq \sum_{i=1}^{n} \frac{1}{x_i}$$

3.10

Rotation matrix in 2D is given by $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

$$\begin{bmatrix} \cos(30^\circ) & -\sin(30^\circ) \\ \sin(30^\circ) & \cos(30^\circ) \end{bmatrix} = \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{3}/2 & -\frac{1}{2} \\ 1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix} + \begin{bmatrix} -\frac{3}{2} \\ \frac{3\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} \sqrt{3} - \frac{3}{2} \\ 1 + \frac{3\sqrt{3}}{2} \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{3}/2 & -\frac{1}{2} \\ 1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ -\sqrt{3}/2 \end{bmatrix}$$